

Q1. The postulates of Quantum Mechanics form the fundamental rules that describe how quantum systems behave.

Postulate 1: The State of a System

Statement:

Statement:

The state of a quantum system is completely described by a *state vector* $|\psi(t)\rangle$ in a *Hilbert space* $\mathcal{H}$. This vector contains all information about the system. The state vector must be **normalized**, such that:

$$\langle\psi|\psi\rangle = 1$$

For a single **qubit**, the state is written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $\alpha, \beta \in \mathbb{C}$, and $|\alpha|^2 + |\beta|^2 = 1$.

Relation to Quantum Computing:

This postulate defines what a **qubit is**. A qubit is the basic unit of quantum information represented as a state in a 2^n dimensional complex Hilbert space of an n -qubit register.

Postulate 2: Observable Quantities Represented by Operators

Statement:

To every dynamical variable A that is a physically measurable quantity, there corresponds an operator A . The operator A is Hermitian and its eigenvectors form a complete orthonormal basis of the vector space.

Using spectral decomposition, we can write the operator A in terms of its eigenvalues and corresponding projection operators $P_n = |u_n\rangle\langle u_n|$ as $A = \sum_n a_n P_n$.

Using spectral decomposition:

$$A = \sum_n a_n P_n$$

where $P_n = |u_n\rangle\langle u_n|$ are projection operators.

Relation to Quantum Computing:

Quantum gates and measurements are represented by operators. Measurement of qubits corresponds to applying Hermitian operators whose outcomes give classical bits (0 or 1).

Explanation:

Hermitian operators guarantee real eigenvalues (since physical measurement results are real).

Examples: Energy $\rightarrow$ Hamiltonian H , Spin $\rightarrow$ Pauli matrices $\sigma_x, \sigma_y, \sigma_z$. The operator encapsulates how measurements act on quantum states.

Postulate 3: Measurements

The possible results of measurement of a dynamical variable A are the eigenvalues a_n of the operator A corresponding to that variable. So, the probability of obtaining a measurement result a_n is

$$\Pr(a_n) = \langle \psi | P_n | \psi \rangle = \text{Tr}(P_n |\psi\rangle\langle\psi|)$$

The probability amplitude $c_n = \langle u_n | \psi \rangle$ gives us,

$$\Pr(a_n) = \frac{|\langle u_n | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|c_n|^2}{\langle \psi | \psi \rangle}$$

After measurement, the state **collapses** to the corresponding eigenstate:

$$|\psi\rangle \xrightarrow{\text{measurement}} \frac{P_n |\psi\rangle}{\sqrt{\langle \psi | P_n | \psi \rangle}}$$

Statement:

When an observable $\hat{A}$ is measured, the only possible outcomes are its eigenvalues a_n . The probability of obtaining outcome a_n is:

$$P(a_n) = \langle \psi | P_n | \psi \rangle = |\langle u_n | \psi \rangle|^2$$

where $P_n = |u_n\rangle\langle u_n|$ is the projection operator onto the eigenspace of a_n .

After measurement, the system *collapses* into the corresponding eigenstate:

$$|\psi\rangle \longrightarrow \frac{P_n |\psi\rangle}{\sqrt{\langle \psi | P_n | \psi \rangle}}$$

Relation :

Measurement plays a key role in probabilistic quantum algorithms such as Grover's algorithm, Shor's algorithm. Measurement extracts the **classical output** of a quantum computation. It causes the superposition of states to collapse to a definite classical bitstring.

Postulate 4: Time Evolution of the System

The time evolution of a closed (i.e., physically isolated) quantum system is governed by the Schrödinger equation. This equation is given by

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H |\psi\rangle$$

where H is the **Hamiltonian operator** representing the system's total energy

The state of the system at a later time t is given by

$$|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$$

Therefore the time evolution of a quantum state is governed by the unitary operator

$$U = e^{-iHt/\hbar}$$

The time evolution of an isolated (closed) quantum system follows the **Schrödinger equation**:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle$$

where H is the **Hamiltonian operator** (total energy of the system).

The state at time t is:

$$|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$$

The time evolution is **unitary**, governed by:

$$U(t) = e^{-iHt/\hbar}$$

Relation:

Quantum gates are **unitary operators**, which evolve quantum states. The Hamiltonian determines how qubits change during computation. Time evolution ensures **reversibility** — an essential property of quantum gates.

Q2. Define a composite quantum system. How is the tensor product used to represent combined states?

In quantum mechanics, many systems involve more than one particle. To describe such multi-particle systems, we construct a composite quantum system

A composite quantum system is a system that consists of two or more subsystems, each with its own Hilbert space. When these systems are combined, the total system is described by a **larger Hilbert space** obtained using the **tensor product** of the individual spaces.

If two subsystems have Hilbert spaces H_1 and H_2 , then the composite system has the Hilbert space:

$$H = H_1 \otimes H_2$$

If the dimensions of these spaces are $\dim(H_1)=N_1$ and $\dim(H_2)=N_2$, then the total space has dimension:

$$\dim(H) = N_1 N_2$$

Representing Combined States using tensor product:

If $|\phi\rangle \in H_1$ and $|\chi\rangle \in H_2$, then the state of the composite system is:

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$$

This tensor product represents a new state vector that lives in the **combined Hilbert space**.

Properties of Tensor Product:

1. Linearity:

$$|\phi\rangle \otimes (|\chi_1\rangle + |\chi_2\rangle) = |\phi\rangle|\chi_1\rangle + |\phi\rangle|\chi_2\rangle$$

2. Scalar multiplication:

$$|\phi\rangle \otimes (\alpha|\chi\rangle) = \alpha(|\phi\rangle \otimes |\chi\rangle)$$

3. Constructing Basis of Composite Space:

If $\{|u_i\rangle\}$ is a basis of H_1 and $\{|v_j\rangle\}$ is a basis of H_2 , then the basis of $H_1 \otimes H_2$ is:

$$|w_{ij}\rangle = |u_i\rangle \otimes |v_j\rangle$$

Often, we write the tensor product without the $\otimes$ symbol, e.g., $|\phi\rangle|\chi\rangle$ or $|\phi\chi\rangle$.

Example (Two Qubits)

Each qubit has a 2-dimensional Hilbert space spanned by $\{|0\rangle, |1\rangle\}$.

For two qubits:

$$H = H_1 \otimes H_2$$

the composite basis is:

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

which corresponds to all tensor products of the single-qubit basis states.

Q3. State the first postulate of quantum mechanics and explain how it relates to the representation of a qubit state in quantum computing. **Q.1....**

Q4. Explain the third postulate of quantum mechanics. What are measurement operators and how do they relate to probabilities? **Q.1....**

Q5 Show that if A and B are Hermitian, then $A \otimes B$ is Hermitian.

Example 4.10

Q6. Calculate the matrix representation of $X \otimes Y$.

1. Recall the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

2. Definition of the tensor (Kronecker) product

For matrices $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ and B ,

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix}.$$

3. Compute $X \otimes Y$

$$X \otimes Y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 \cdot Y & 1 \cdot Y \\ 1 \cdot Y & 0 \cdot Y \end{pmatrix} = \begin{pmatrix} 0 & Y \\ Y & 0 \end{pmatrix}.$$

Now substitute Y :

$$X \otimes Y = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}.$$

Q7 Calculate the matrix representation of $Z \otimes X$, showing that $X \otimes Z \neg Z \otimes X$.

Step 1: Define the Pauli matrices

The Pauli-Z and Pauli-X matrices are 2×2 matrices defined as:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Step 2: Calculate the tensor product $Z \otimes X$

The tensor product (Kronecker product) of two matrices, A and B, is given by the formula:

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix}$$

Using this, we can calculate $Z \otimes X$:

$$Z \otimes X = \begin{pmatrix} 1 \cdot X & 0 \cdot X \\ 0 \cdot X & -1 \cdot X \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \end{pmatrix}$$
$$Z \otimes X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

Step 3: Calculate the tensor product $X \otimes Z$

Next, we calculate the tensor product $X \otimes Z$:

$$X \otimes Z = \begin{pmatrix} 0 \cdot Z & 1 \cdot Z \\ 1 \cdot Z & 0 \cdot Z \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix}$$
$$X \otimes Z = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

Answer:

The matrix representation of $Z \otimes X$ is:

$$Z \otimes X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

The matrix representation of $X \otimes Z$ is:

$$X \otimes Z = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

Q8 Define the density operator for a pure state $|\phi\rangle$. Show that $\rho^2 = \rho$ in this case.

In quantum mechanics, the density operator (or density matrix) is used to describe the statistical state of a quantum system. For a **pure state** $|\phi\rangle$, the density operator is defined as the **outer product** of the state vector with itself which is represented as

$$\rho = |\phi\rangle\langle\phi|$$

It contains complete information about the quantum state.

Properties of a Density Operator for a Pure State

1. The density operator is Hermitian, meaning $\rho^\dagger = \rho$
2. Trace $\text{Tr}(\rho) = 1$
3. For a pure state only, the density operator is idempotent, meaning $\rho^2 = \rho$

Showing that $\rho^2 = \rho$ (Idempotent Property)

Given:

$$\rho = |\phi\rangle\langle\phi|$$

Compute ρ^2 :

$$\rho^2 = (|\phi\rangle\langle\phi|)(|\phi\rangle\langle\phi|)$$

Group the middle terms:

$$\rho^2 = |\phi\rangle(\langle\phi|\phi\rangle)\langle\phi|$$

Since the state is normalized, we know:

$$\langle\phi|\phi\rangle = 1$$

Substitute:

$$\rho^2 = |\phi\rangle(1)\langle\phi|$$

$$\rho^2 = |\phi\rangle\langle\phi|$$

$$\rho^2 = \rho$$

Q9. Define the density operator for a mixed state. Give an example with $\rho = 0.6|0\rangle\langle 0| + 0.4|1\rangle\langle 1|$ and explain its interpretation.

When a quantum system is not in a definite single state but rather in a **statistical mixture** of several possible pure states, we describe it using a **mixed state density operator**.

The **density operator for a mixed state** is defined as:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where:

- $|\psi_i\rangle$ are normalized pure states,
- p_i are the probabilities of finding the system in each state, satisfying $\sum_i p_i = 1$.

This form expresses the fact that we do **not know precisely** which pure state the system is in, only the **probability distribution** over possible states.

Example:

Given:

$$\rho = 0.6|0\rangle\langle 0| + 0.4|1\rangle\langle 1|$$

Represent the projectors as matrices:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Now substitute:

$$\rho = 0.6 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0.4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$
$$\rho = \begin{bmatrix} 0.6 & 0 \\ 0 & 0.4 \end{bmatrix}$$

Interpretation

1. The system has a **60% probability** of being in the state $|0\rangle$ and a **40% probability** of being in the state $|1\rangle$.
2. Because there are no off-diagonal terms, the state has **no quantum coherence** — meaning it is just a **classical mixture**, not a quantum superposition.
3. The **trace** of the density operator gives the total probability:

$$\text{Tr}(\rho) = 0.6 + 0.4 = 1$$

which confirms correct normalization.

4. The **purity** of the state can be checked by:

$$\text{Tr}(\rho^2) = (0.6)^2 + (0.4)^2 = 0.52 < 1$$

Since $\text{Tr}(\rho^2) < 1$, it confirms that the state is **mixed**, not pure.

Q10. Express the mixed state $\rho = 0.8|0\rangle\langle 0| + 0.2|1\rangle\langle 1|$ on the Bloch sphere and compute its Bloch vector $\vec{r}$.

1. Recall the general form

For any single-qubit state, the **density operator** can be written in the **Bloch sphere representation** as:

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$$

where:

- I is the 2×2 identity matrix,
- $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices,
- $\vec{r} = (r_x, r_y, r_z)$ is the **Bloch vector** with magnitude $|\vec{r}| \leq 1$.

For **pure states**, $|\vec{r}| = 1$;

for **mixed states**, $|\vec{r}| < 1$.

2. Write ρ in matrix form

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

So:

$$\rho = 0.8 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 0.2 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.8 & 0 \\ 0 & 0.2 \end{pmatrix}.$$

3. Compare with Bloch form

The general Bloch form is:

$$\rho = \frac{1}{2} \begin{pmatrix} 1 + r_z & r_x - ir_y \\ r_x + ir_y & 1 - r_z \end{pmatrix}.$$

From our matrix:

$$\rho = \begin{pmatrix} 0.8 & 0 \\ 0 & 0.2 \end{pmatrix}.$$

By comparing entries:

$$\frac{1}{2}(1 + r_z) = 0.8, \quad \frac{1}{2}(1 - r_z) = 0.2, \quad r_x = 0, \quad r_y = 0.$$

4. Solve for r_z

From the first equation:

$$\frac{1 + r_z}{2} = 0.8 \Rightarrow 1 + r_z = 1.6 \Rightarrow r_z = 0.6.$$

Pauli matrices (for reference)

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The Bloch vector components are $r_i = \text{Tr}(\rho \sigma_i)$.

Compute the Bloch components

- $r_x = \text{Tr}(\rho \sigma_x) = \text{Tr} \begin{pmatrix} 0.8 & 0 \\ 0 & 0.2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = 0.$
- $r_y = \text{Tr}(\rho \sigma_y) = 0$ (off-diagonal elements of ρ are zero).
- $r_z = \text{Tr}(\rho \sigma_z) = 0.8(1) + 0.2(-1) = 0.8 - 0.2 = 0.6.$

So

$$\vec{r} = (r_x, r_y, r_z) = (0, 0, 0.6).$$

OR

Bloch-sphere representation & interpretation

Write $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$:

$$\rho = \frac{1}{2}(I + 0.6\sigma_z) = \frac{1}{2} \begin{pmatrix} 1 + 0.6 & 0 \\ 0 & 1 - 0.6 \end{pmatrix} = \begin{pmatrix} 0.8 & 0 \\ 0 & 0.2 \end{pmatrix},$$

Interpretation on the Bloch Sphere

>>> Geometrically: the state corresponds to the point on the Bloch sphere **along the +z axis** (toward $|0\rangle$, the north pole) at distance $|\vec{r}|=0.6$ from the origin.

>>> Magnitude: $|\vec{r}|=0.6 < 1$, so the state is **mixed** (pure states have $|\vec{r}|=1$).

>>> A pure $|0\rangle$ state would have $\vec{r}=(0,0,1)$, and a pure $|1\rangle$ would have $\vec{r}=(0,0,-1)$.

This mixed state lies **between the centre and the north pole** of the Bloch sphere.

Q11. Consider the following matrix:

Is this a valid density operator?

$$p = \begin{pmatrix} \frac{5}{8} & \frac{i}{4} \\ \frac{-i}{4} & \frac{3}{8} \end{pmatrix}$$

Q12. Consider the following matrix:

Does this represent a pure state or a mixed state?

$$p = \begin{pmatrix} \frac{5}{8} & \frac{i}{4} \\ \frac{-i}{4} & \frac{3}{8} \end{pmatrix}$$

Example 5.12

Q13 Explain completely mixed states.

Completely Mixed States

A completely mixed state can be thought of as the opposite end along a continuum of density operators, with a pure state on the other side. In a completely mixed state the probability for the system to be in each given state is identical. In that case the density operator is a constant multiple of the identity matrix. If the state space has n dimensions, then

$$\rho = \frac{1}{n} I \quad (5.13)$$

Since $I^2 = I$, we have $\rho^2 = \frac{1}{n^2} I$. Furthermore, in n dimensions, $\text{Tr}(I) = n$. So for a completely mixed state we have

$$\text{Tr}(\rho^2) = \text{Tr}\left(\frac{1}{n^2} I\right) = \frac{1}{n^2} \text{Tr}(I) = \frac{1}{n} \quad (5.14)$$

In most if not all cases of interest to us, $n = 2$. For $n = 2$ the lower bound, given by a completely mixed state, is $\text{Tr}(\rho^2) = \frac{1}{2}$ while the upper bound for a pure state is given by $\text{Tr}(\rho^2) = 1$.

Formula idhar se chaapna hai aur trace ka explanation agar dena ho toh bhi

2. Physical Meaning

The completely mixed state represents a situation where:

- The system is **equally likely** to be found in any of its possible pure states.
- We have **maximum ignorance** (no knowledge) about the actual pure state.

In contrast:

- A **pure state** (e.g., $|\psi\rangle$) represents **complete knowledge** of the system, with $\rho = |\psi\rangle\langle\psi|$ and $\text{Tr}(\rho^2) = 1$.
- A **completely mixed state** represents **no knowledge**, with $\text{Tr}(\rho^2) = \frac{1}{N}$.

3. Example: One Qubit

A single qubit has a 2-dimensional Hilbert space ($N = 2$).

The completely mixed state is:

$$\rho = \frac{1}{2} I = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

This corresponds to being in $|0\rangle$ or $|1\rangle$ with equal probability $\frac{1}{2}$, and with **no coherence** (off-diagonal terms = 0).



A **completely mixed state** (also called a **maximally mixed state**) is a quantum state that represents **maximum uncertainty** about the state of a system — it contains **no information** about which pure state the system is in.

It is the **quantum analogue** of a classical uniform probability distribution over all possible outcomes.

1. Definition

For a quantum system with a Hilbert space of dimension N , the **completely mixed state** is described by the **density operator**

$$\rho = \frac{1}{N} I$$

where:

- I is the $N \times N$ identity matrix, and
- $\text{Tr}(\rho) = 1$ ensures proper normalization.

This density matrix corresponds to an equal probabilistic mixture of all orthonormal basis states.



$$p = \begin{pmatrix} \frac{5}{8} & \frac{i}{4} \\ -\frac{i}{4} & \frac{3}{8} \end{pmatrix}$$

14) Suppose matrix

Show that p is Hermitian and has positive eigenvalues that satisfy $0 \leq \lambda_i \leq 1$, and $\sum \lambda_i = 1$.

Similar to Example 5.9

$$p = \begin{pmatrix} \frac{5}{8} & \frac{i}{4} \\ -\frac{i}{4} & \frac{3}{8} \end{pmatrix}$$

15) Suppose

Is this a mixed state?

Find $\langle X \rangle$ for this state.

Similar to Example 5.9

(Yeh sir ko puchna padega kyuki matrix alag hai textbook me toh use galti nahi hui na to example wo wala bhi karna padega)